

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B. Tech (CBCS) Degree Examinations

Summer Semester July 2024
RI2006-1 – INTRODUCTION TO ROBOTICS
(RI)

Duration: 3 Hours

Max. Marks: 100

Note: Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- What does the abbreviation SCARA stand for in the context of robotics
 A) Simple Circular Arm Robot Algorithm B) Standard Configurable Autonomous Robotic Arm
 C) Simple Cartesian Arm Robotics Application D) Selective Compliance Assembly Robot Arm
- What does the work volume of a robot refer to?
 A) The robot's weight B) The storage capacity of the robot
 C) The sphere within which the robot can manipulate objects D) The energy consumption of the robot
- Which factor is NOT generally considered when choosing a robot for a specific application?
 A) Number of axes B) Repeatability
 C) Work volume D) Color
- What does DOF stand for in robotics?
 A) Direction of Force B) Degrees of Friction
 C) Dynamic of Frame D) Degrees of Freedom
- Given a point $a_{uvw} = (4, 3, 2)^T$ with respect to the rotated OUVW coordinate system, determine corresponding point a_{xyz} with respect to the reference coordinate system if it has been rotated 60° about OZ axis.
 A) $\begin{bmatrix} 0.598 \\ 4.964 \\ -2 \end{bmatrix}$ B) $\begin{bmatrix} -0.598 \\ 4.964 \\ 2 \end{bmatrix}$
 C) $\begin{bmatrix} 4 \\ 1.26 \\ -2 \end{bmatrix}$ D) $\begin{bmatrix} -4 \\ -1.26 \\ -2 \end{bmatrix}$
- In what situation would adhesive grippers generally NOT be suitable?
 A) Handling porous materials B) Handling glass surfaces
 C) Handling metal sheets D) Handling plastic parts
- If $a_{xyz} = (6, 2, 4)^T$ are the coordinates with respect to the reference coordinate system, determine the corresponding points a_{uvw} with respect to the rotated OUVW coordinate system if it has been rotated 60° about the OZ axis.
 A) $\begin{bmatrix} 4.732 \\ -4.196 \\ 4 \end{bmatrix}$ B) $\begin{bmatrix} -4.732 \\ 4.196 \\ 4 \end{bmatrix}$
 C) $\begin{bmatrix} 3.738 \\ 2.196 \\ 2 \end{bmatrix}$ D) $\begin{bmatrix} 3.738 \\ -2.196 \\ 2 \end{bmatrix}$
- A point $a_{uvw} = (2, 3, 4)$ is to be translated a distance -6 units along the OY axis and -4 units along the OZ axis. Using the appropriate homogeneous transformation matrix, determine the new point a_{xyz} .
 A) $\begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \end{bmatrix}$ B) $\begin{bmatrix} -2 \\ 3 \\ 0 \\ 1 \end{bmatrix}$
 C) $\begin{bmatrix} -3 \\ -2 \\ -1 \\ 1 \end{bmatrix}$ D) $\begin{bmatrix} 3 \\ -2 \\ 1 \\ 1 \end{bmatrix}$

9. What is LVDT
 A) Linear variable differential transistor B) Low Voltage Differential Transistor
 C) Linear Voltage Digital Transformer D) Linear variable Differential Transformer
10. Which of the following is an advantage of using pneumatic actuators?
 A) High precision B) Simplicity and reliability
 C) Energy efficiency D) High torque output
11. What is the primary function of an optical encoder?
 A) Temperature measurement B) Pressure monitoring
 C) Position and speed sensing D) Voltage regulation
12. What type of sensor generates its own energy to collect data?
 A) Active sensor B) Passive Sensor
 C) Infrared sensor D) Ultrasonic sensor
13. Robotic exoskeletons are primarily used in:
 A) Medical Imaging B) Rehabilitation
 C) Remote Patient Monitoring D) Telemedicine
14. What component is commonly found in a brushed DC motor but not in a brushless DC motor?
 A) Stator B) Commutator
 C) Rotor D) Permanent magnet
15. Which application is most suitable for series DC motors?
 A) High-speed, low-torque applications B) High-torque, low-speed applications
 C) Long-duration, continuous usage D) Precision control
16. What distinguishes a compound DC motor from other types?
 A) It uses permanent magnets B) It does not have a commutator
 C) It operates at high speeds D) It has both series and shunt windings
17. Which application of robotics aids in helping patients recover mobility and strength?
 A) Telepresence Robots B) Surgical Robots
 C) Rehabilitation D) Service Robots
18. What is one reason Underwater Gliders are energy-efficient?
 A) They use nuclear power B) They use changes in buoyancy to move
 C) They use gasoline engines D) They are wind-powered
19. AI in healthcare is most helpful in which of the following contexts?
 A) Writing prescriptions B) Organ Transplant
 C) Medical Imaging D) Record-keeping
20. Which robots are designed to provide companionship and emotional support?
 A) Surgical Robots B) Telepresence Robots
 C) Social Robots D) Rehabilitation Robots

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit - I**

Marks BT* CO* PO*

1. a) A cylindrical robot has a prismatic joint with a range of travel of 800mm. The control memory for this joint has 10bit capacity. It has been recorded that the associated mechanical inaccuracy with the said arm show a random distribution of random variable of the robot position about the mean position of the taught point gives a standard deviation of 0.1mm. The standard deviation is equal in all direction. Determine the following,
 I. The control resolution for the axis.
 II. The accuracy defined.
 III. The repeatability of the robot link.
- b) There is a reference coordinate frame {1} and moving coordinate frame {2} which are coinciding initially. The coordinates of point P in frame {2} are $[3 \ 2 \ 1]^T$. The moving coordinate frame {2} is now rotated about z-axis of the reference coordinate frame {1} by 45 degrees. Find the coordinates of point P with respect to the reference coordinate frame {1} after this rotation.

8 L*2 1 2

8 L2 2 2

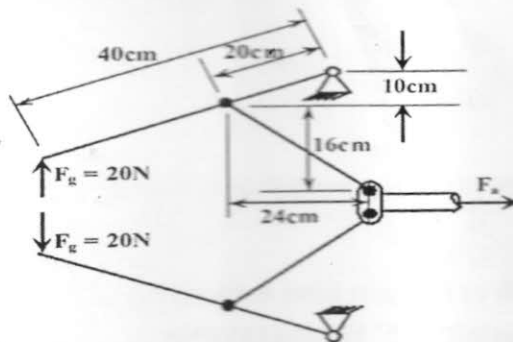
- 8 L2 1 1

-

8 L3 2 2

- 4 L2 1 1

- 4 L2 1 1



8 L3 2 2

8 L1 3 1

- 8 L2 4 1

- | | | | | | |
|-------|---|---|----|---|---|
| 5. a) | Explain time-of-flight method and Structured light approach method of range sensing. | 8 | L1 | 3 | 2 |
| b) | What are Induction motors? How are they classified? Explain the working of a Single-phase Squirrel-cage Induction Motor with a neat sketch. | 8 | L1 | 4 | 1 |
| 6. a) | What are the dynamic characteristics one should check while selecting a sensor for an application? Explain. | 8 | L1 | 3 | 1 |
| b) | Sketch and explain working principle and control of brushless DC Motor with switching circuit. | 8 | L1 | 4 | 1 |

Unit – III

- | | | | | | |
|-------|---|---|----|---|---|
| 7. a) | What is Rehabilitation Robotics? Explain its types? | 8 | L1 | 5 | 1 |
| b) | What are Aerial Robots? Explain different types of Unmanned Aerial Vehicles. | 8 | L1 | 5 | 1 |
| 8. a) | How are Robots changing Healthcare? Explain. | 8 | L1 | 5 | 1 |
| b) | What are the types of Underwater Robots? What are the challenges encountered in developing Underwater Robots? | 8 | L1 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
